组会报告

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2020年4月9日

Motivation

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.

提升建模长距离依赖的方法：

1. 自注意力/non-local，缺点是对每个空间位置计算仿射矩阵消耗大量内存。
2. 空洞卷积或全局/金字塔池化，缺点是都只在正方形窗口内探测输入特征。限制了他们捕捉异形上下文的灵活性。会引入不相关的区域特征。

解决方法：提出 Strip Pooling

优点：

1. 在一个方向上采用狭长的核，捕捉孤立区域的长距离关系。2. 另一方向上也采用狭长的核，捕捉局部上下文，避免不相关区域的干扰。

Motivation

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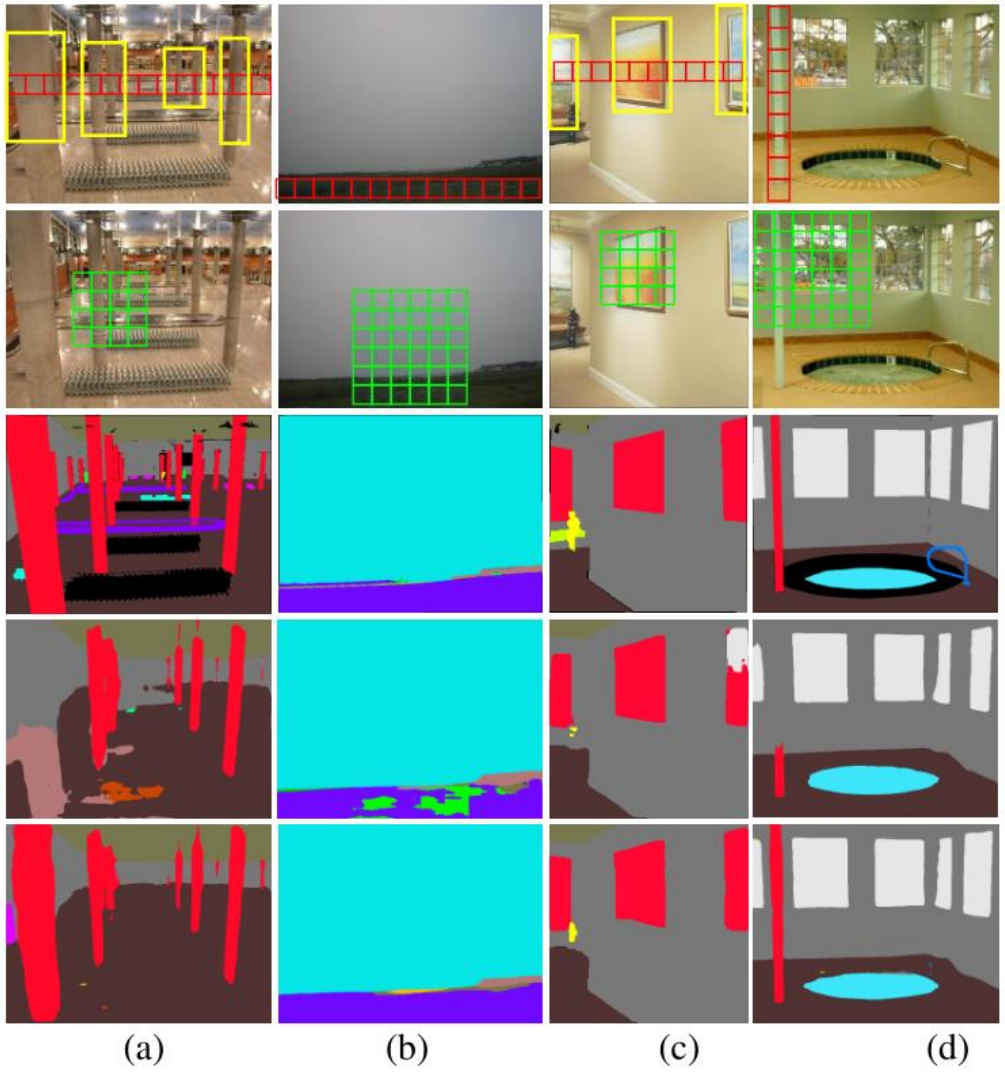


Strip Pooling 和 Spatial Pooling 对比

Method

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.

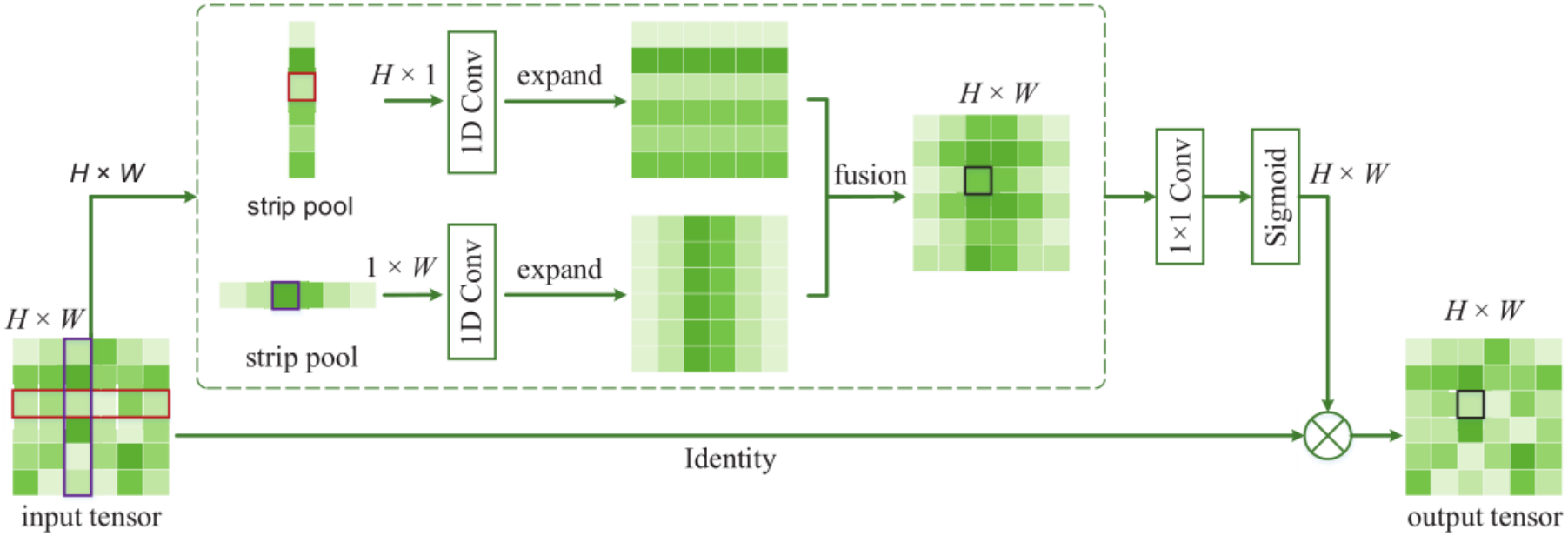


Figure 2. Schematic illustration of the Strip Pooling (SP) module.

$$y_i^h = \frac{1}{W} \sum_{0 \le j < W} x_{i,j}.$$

水平池化

$$y_j^v = \frac{1}{H} \sum_{0 \le i < H} x_{i,j}.$$

垂直池化

Strip Pooling 模块，用于 Backbone 中

Method

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.

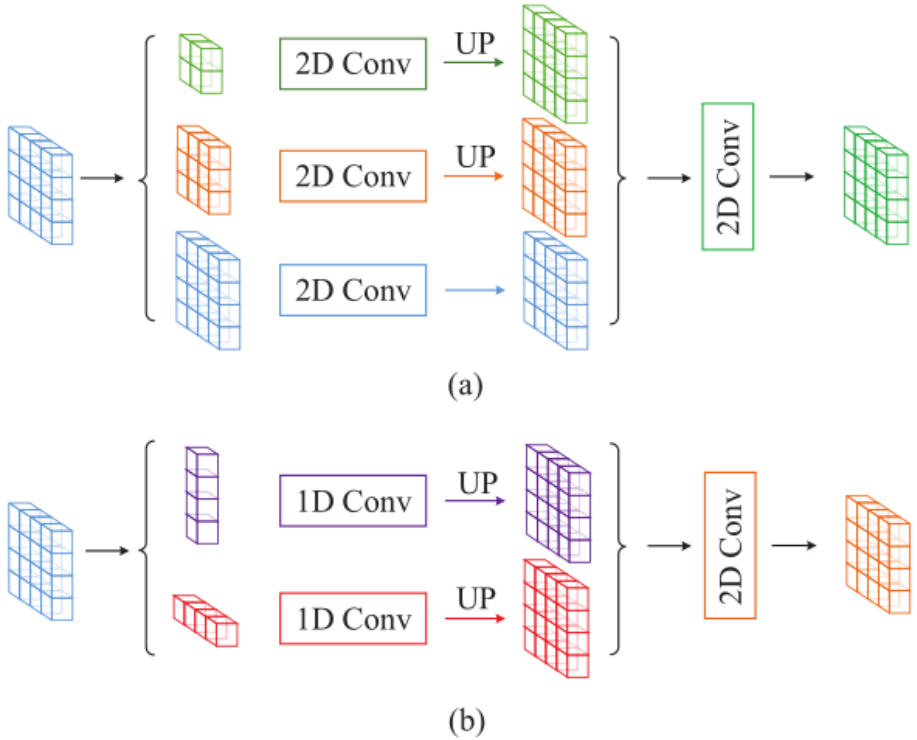


Figure 3. (a) Short-range dependency aggregation sub-module. (b) Long-range dependency aggregation sub-module. Inspired by [34, 35], a convolutional layer is added after the fusion operation in each sub-module to reduce the aliasing effect brought by down-sampling operations.

Mixed Pooling 模块，用于 Backbone 之后

Experiments

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.

Settings	#Params	SPM	mIoU	Pixel Acc
Base FCN	27.7 M	✗	37.63	77.60%
Base FCN + PPM [65]	+21.0 M	✗	41.68	80.04%
Base FCN + 1 MPM	+4.4 M	✗	40.50	79.60%
Base FCN + 2 MPM	+8.8 M	✗	41.92	80.03%
Base FCN + 2 MPM	+11.9 M	✓	**44.03**	**80.65%**

Table 1. Ablation analysis on the number of mixed pooling modules (MPMs). 'SPM' refers to the strip pooling module. As can be seen, when more MPMs are used, better results are yielded. All results are based on ResNet-50 backbone and single-model test. Best result is highlighted in **bold**.

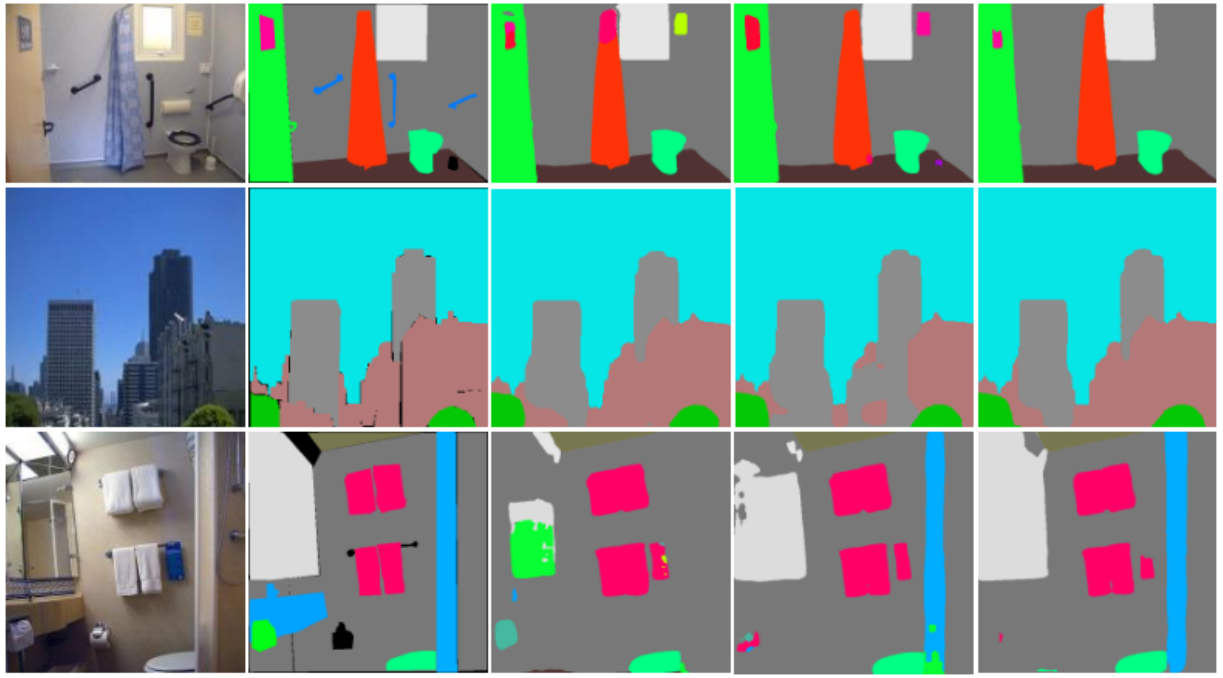
Strip Pooling 和 Mixed Pooling 实验对比

Experiments

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.

Settings	w/ SPM	mIoU	Pixel Acc
Base FCN	✗	37.63	77.60%
Base FCN + 2 MPM (SRD only)	✗	40.50	79.34%
Base FCN + 2 MPM (LRD only)	✗	41.14	79.64%
Base FCN + 2 MPM (SRD + LRD)	✗	41.92	80.03%
Base FCN + 2 MPM (SRD + LRD)	✓	44.03	80.65%

Table 2. Ablation analysis on the mixed pooling module (MPM). 'SPM' refers to the strip pooling module. 'SRD' and 'LRD' denote the short-range dependency aggregation sub-module and the long-range dependency aggregation sub-module, respectively. As can be seen, collecting both short-range and long-range dependencies are essential for yielding better segmentation results. All results are based on single-model test.



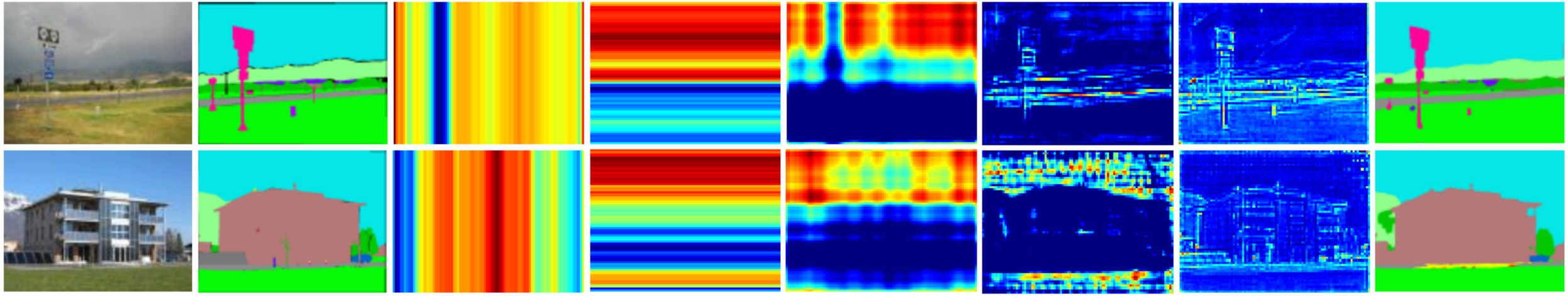
(a) Image (b) GT (c) 2 SRD (d) 2 LRD (e) 2 MPM

Figure 4. Visual comparisons among different settings of the MP module (MPM). '2 SRD' means we use 2 MPMs with only the short-range dependency aggregation module included and '2 LRD' means we use 2 MPMs with only the long-range dependency aggregation module included.

短距离依赖/长距离依赖实验对比

Experiments

Hou, Q., et al. (2020). Strip Pooling: Rethinking Spatial Pooling for Scene Parsing. CVPR2020.



(a) Image (b) GT (c) After VSP (d) After HSP (e) After LRD (f) After SRD (g) After MPM (h) Results

Figure 5. Visualization of selected feature maps at different positions of the proposed MP module. **VSP**: vertical strip pooling; **HSP**: horizontal strip pooling; **SRD**: short-range dependency aggregation sub-module (Figure 3a); **LRD**: long-range dependency aggregation sub-module (Figure 3b); **MPM**: mixed pooling module.

Mixed Pooling 模块特征图可视化